

DESi ablation Phase-4 — automatic state construction, neural retrieval, 8 cases × 3 models

Real run · temp 0 · 2 reps · 8 lifecycle/supersession cases · Sonnet-4.5 / GPT-4o / Llama-3.3-70B. B_auto = state built by ONE LLM extraction pass over the raw chat (told to drop reversed/ruled-out items); R2n = neural-embedding retrieval (bge-small). Conservative; corrects the Phase-3 magnitude.

Mean recall (8 cases) per condition × model

model	A	B	B_auto	R1 bm25	R2 tfidf	R2n neural	E
Sonnet-4.5	0.938	1.0	0.95	0.537	0.562	0.85	0.988
GPT-4o	0.85	0.988	0.95	0.488	0.45	0.713	1.0
Llama-3.3-70B	0.725	1.0	0.875	0.513	0.488	0.725	0.975

Key deltas (mean ± sd over 8 cases; N≈ to detect 0.1)

comparison	note	Sonnet-4.5	GPT-4o	Llama-3.3-70B
B – B_auto	auto-construction works?	+0.05±0.093 (N14)	+0.04±0.106 (N18)	+0.12±0.104 (N17)
B – R1 bm25	vs lexical retrieval	+0.46±0.16 (N41)	+0.50±0.233 (N86)	+0.49±0.181 (N52)
B – R2n neural	vs NEURAL retrieval	+0.15±0.141 (N32)	+0.28±0.198 (N62)	+0.28±0.219 (N76)
B_auto – R2n neural	auto-state vs neural retrieval	+0.10±0.107 (N18)	+0.24±0.192 (N58)	+0.15±0.262 (N108)
B – A	vs full chat	+0.06±0.092 (N14)	+0.14±0.213 (N72)	+0.28±0.158 (N40)
B – E	governance (metadata)	+0.01±0.035 (N2)	-0.01±0.035 (N2)	+0.03±0.071 (N8)

confidence_while_wrong rate (confidently reporting stale content)

model	R1 bm25	R2n neural	B	B_auto
Sonnet-4.5	0.25	0.0	0.0	0.0
GPT-4o	0.5	0.125	0.0	0.0
Llama-3.3-70B	0.5	0.25	0.0	0.0

Conservative findings

1. Automatic state construction WORKS — this addresses the Phase-3 caveat. One LLM extraction pass that resolves supersession yields B_auto ≈ B (curated): B–B_auto = +0.04 to +0.13 (within noise at N=8), worst single case 0.80, across all 3 models. And B_auto beats retrieval (~+0.4 vs lexical, ~+0.15 vs neural). So DESi's value here is CONSTRUCTIBLE, not merely hand-curated ground truth. The structuring helps weaker models most (B–A: Sonnet +0.06, Llama +0.28).

2. Neural retrieval CORRECTS the magnitude (honest walk-back of Phase 3). Phase-3's lexical retrieval (BM25/TF-IDF) overstated DESi's edge (~ 0.7). A proper neural retriever (R2n, bge-small) recovers most of the live content (0.71–0.85), so B–retrieval shrinks to **$\sim 0.15\text{--}0.28$** (modest, borderline at N=8: N $\approx$ 32–76 needed for a 0.1 CI). The residual is supersession-resolution: retrieval surfaces stale AND live passages; B/B_auto carry only the resolved live state. 'DESi beats retrieval' holds, but the honest gap vs STRONG retrieval is small, not large.

3. Replicates across 8 cases $\times$ 3 models. B–R1 $\approx$ +0.46–0.50, B–R2n $\approx$ +0.15–0.28, B–B_auto $\approx$ 0, B–E $\approx$ 0 — consistent on Sonnet, GPT-4o and Llama. The large effects (B–R1, B–B_auto) are well-powered at N=8; B–R2n is borderline (would want $\sim 15\text{--}20$ cases).

4. Confidently-reporting-stale is a RETRIEVAL pathology. confidence_while_wrong: R1 = 0.25–0.5, R2n = 0–0.25, B = B_auto = 0. The resolved state removes the confident-stale failure mode that retrieval (especially lexical) exhibits.

5. Metadata governance still NOT established (B–E $\approx$ 0 on all 3 models, all phases). The live-state SELECTION / supersession-resolution carries the value, not the category/typing metadata.

Net: Supported — correct, AUTO-CONSTRUCTIBLE live state is load-bearing and beats retrieval on lifecycle/supersession tasks (modestly vs neural retrieval; the value is resolving stale content). Not supported — metadata governance; 'DESi makes the model smarter' / 'wins generally'. The Phase-3 'huge gap vs retrieval' was a lexical-retrieval artefact.

Cost: 3 models $\times$ 8 cases $\times$ 7 conditions $\times$ 2 reps + 24 extraction $\approx$ 360 calls. Next: 15–20 cases to tighten the B–R2n CI; a held-out extractor model for B_auto; harder multi-supersession chains.